6.

Exiting gdb Once you have finished debugging, you can exit gdb with the `quit` command:

(gdb) quit

7.

Advanced gdb Features

- **Backtrace:** In case of crashes, you can use `backtrace` to view the function call stack and locate where the error occurred.
- **Watchpoints:** These allow you to monitor a variable's value during execution, pausing when it changes.
- **Conditional Breakpoints:** You can set breakpoints that only activate when a specific condition is true.

Best Practices for Efficient Code

Debugging and optimization are two sides of the same coin. While debugging helps fix errors, optimization ensures that your code runs faster and uses fewer resources. Below are some best practices to keep in mind when optimizing your code:

1. **Write Clean and Readable Code** The best way to reduce bugs is to prevent them in the first place. Write clean, readable, and maintainable code. Use meaningful variable names, keep functions small and focused, and follow a consistent coding style. A clean codebase is much